

R-Studio Climate Data Extraction Guide

This reference guide is specifically created based on the R script file “DataExtraction from large file.R” which extracts climate data from NetCDF files covering the period 1950-2024. However, the exact same process and steps outlined in this guide can be applied to all three R scripts provided in the file directory:

- **DataExtraction from large file.R** - Extracts climate data from 1950-2024 using standard processing
- **DataExtraction from large fil multithread.R** - Uses all CPU cores in the computer for faster data extraction from climate data covering 1950-2024
- **First_data_extraction.R** - Extracts climate data from the shorter period of 2011-2024

Follow the same steps outlined below but open the appropriate R script file based on the data requirements and processing preferences. The modification points are file paths, locations, and NetCDF file names. All other components in the code should remain the same across all three scripts.

Prerequisites

- R-Studio installed on the computer
- NetCDF climate data files (.nc format)
- Basic understanding of file paths on the system

Step 1: Install Required Packages

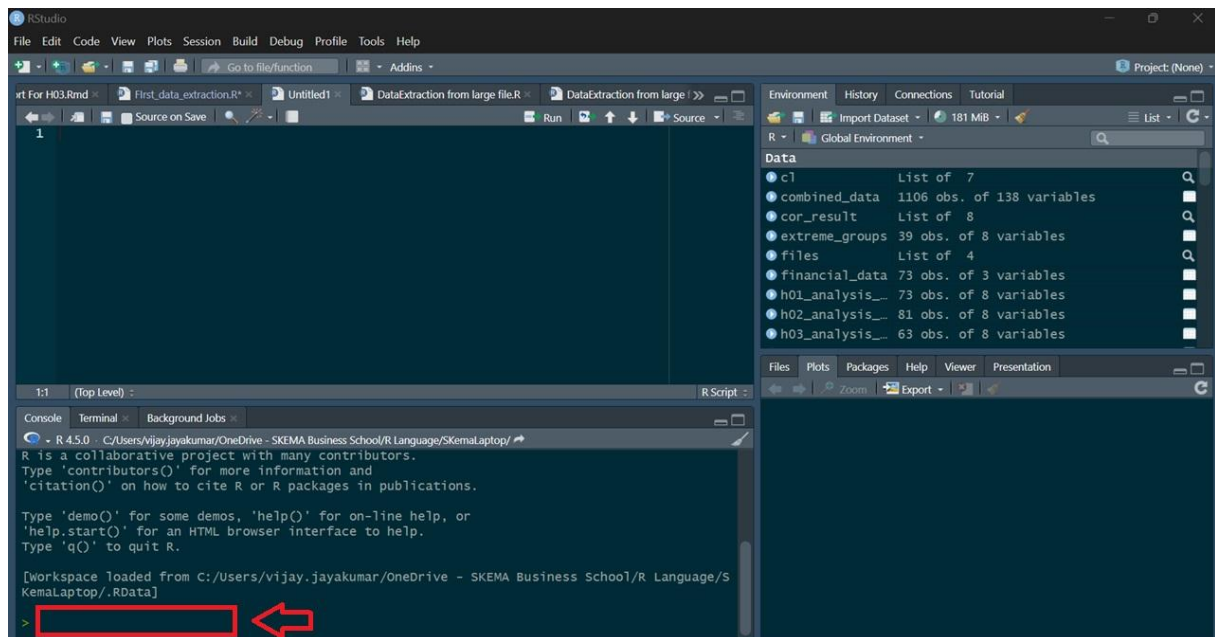
Copy and paste each line below into the R-Studio Console (bottom-left panel) and press Enter after each line. This process is required for the first time and for subsequent run this process is not necessary.

```
# Install required packages (run once)

install.packages("ncdf4")

install.packages("tidyverse")

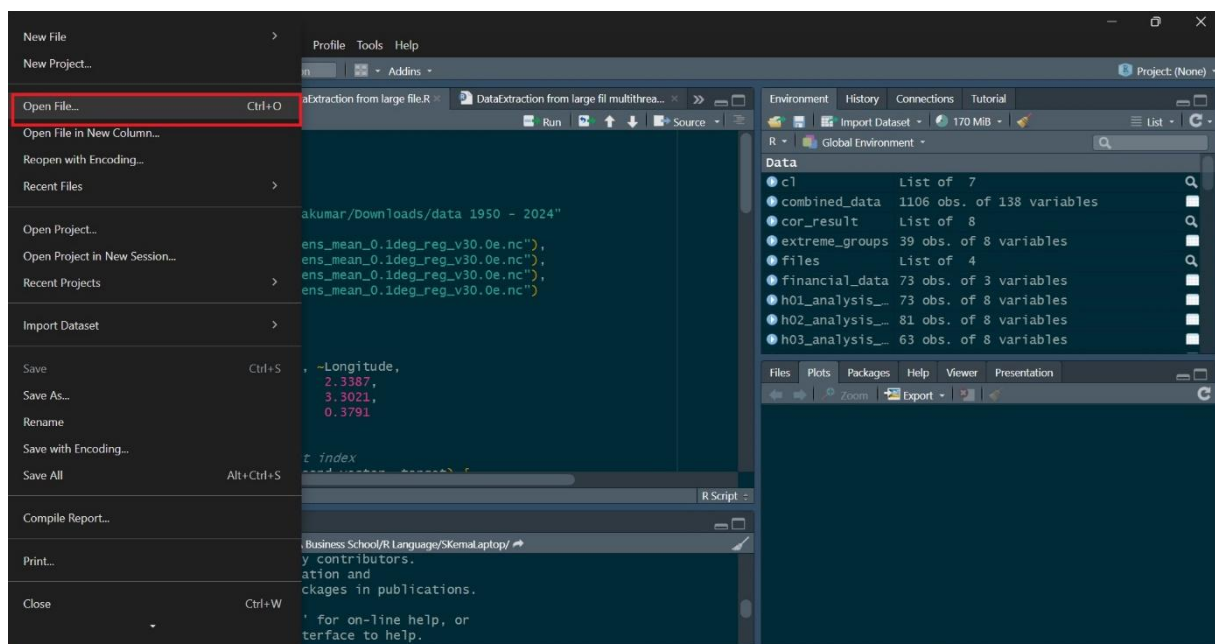
install.packages("lubridate")
```



This Downloads and installs the necessary tools for working with NetCDF files and data manipulation.

Step 2: Using the Complete Script

1. Open R-Studio
2. Go to File → Open File
3. Navigate to the directory and open the file: “DataExtraction from large file.R”
4. Modify the following sections before running:
 - **Line 9:** Change data_dir to the actual data folder path
 - **Lines 18-24:** Update locations with the specific coordinates (up to 30 locations)
 - **Lines 11-16:** Verify the NetCDF file names match exactly
5. Select all code (Ctrl+A) and run it (Ctrl+Enter) to run all code at one go. Alternatively, use (Ctrl+Enter) for each line starting from line 1 till the last active line.



File Structure Expected:

- The R script file: DataExtraction from large file.R
- The NetCDF files in the same directory or specified path
- Output will be saved as: Climate_Data_1950_2024.csv

Always modify the file paths and locations to match the specific needs before running the script.

Steps describing the functionality of each code block in the R-Script

From step 3 to step 9 is provided in this document for reference purpose. This helps to understand each line of code and can identify potential errors that can arise while running the script.

Step 3: Load Required Libraries

Code Reference: Lines 5-7

- # Load libraries (run every session)
- library(ncdf4), library(tidyverse), library(lubridate)

This Activates the tools that are installed.

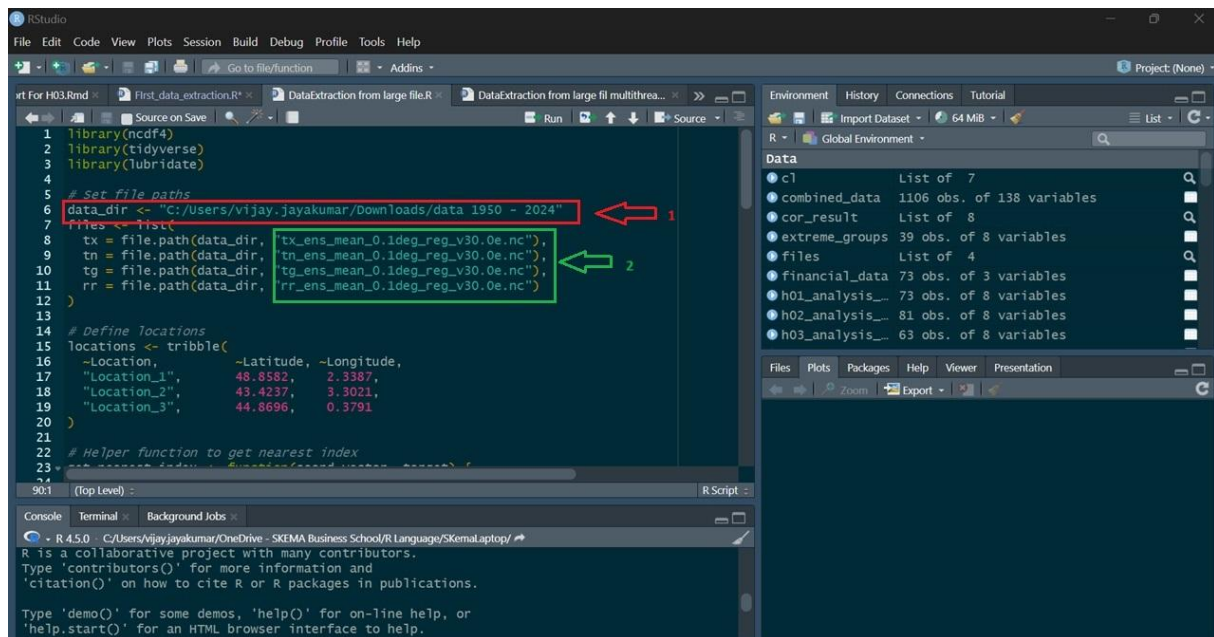
Step 4: Set Up The File Paths

Code Reference: Lines 9-16

- # Set file paths
- data_dir <- "C:/Users/vijay.jayakumar/Downloads/data 1950 - 2024"
- files <- list(...) with comments for each file type

Important Notes:

- Use forward slashes (/) or double backslashes (\\) in file paths
- Make sure the file names match exactly
- Check that all files exist in the specified folder
- The file path mentioned in the screenshot as (1) should be changed and the file name as (2) should not be changed.



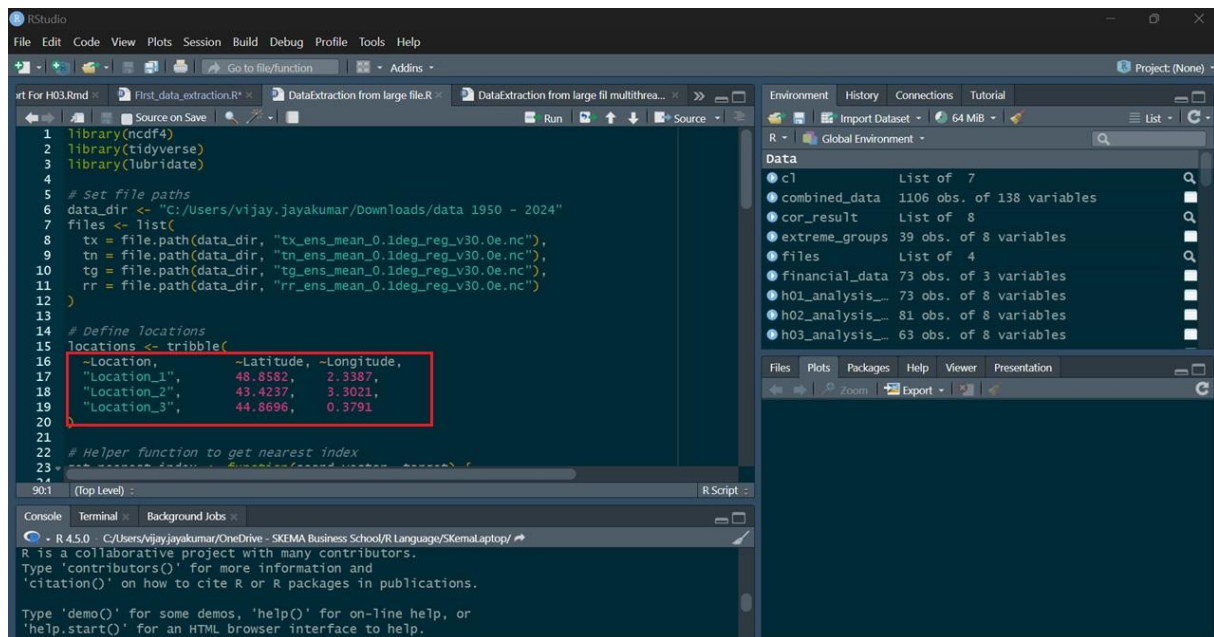
Step 5: Define The Locations (Maximum 30)

Code Reference: Lines 18-24

- # Define locations
- locations <- tribble(~Location, ~Latitude, ~Longitude, ...)

Location Guidelines:

- Latitude: North is positive, South is negative (-90 to +90)
- Longitude: East is positive, West is negative (-180 to +180)
- Use descriptive names without spaces or special characters
- The script contains only 3 locations
- Multiple locations can be added to the script similar to the structure provided in the file. (Syntax: "Location_1", 48.8582, 2.3387,)
- Maximum 30 locations per extraction



Step 6: Set Up Helper Functions

Code Reference: Lines 26-29 and 39-47

- # Helper function to get nearest index
- `get_nearest_index <- function(coord_vector, target) {...}`
- # Function to extract values from a file
- `extract_variable <- function(file, varname, lat_idx, lon_idx) {...}`

This step creates reusable functions that will find the closest data points to the locations and extract the climate data.

Step 7: Read Coordinate and Time Information

Code Reference: Lines 31-37

- # Open one file to read dimensions
- `ncin <- nc_open(files$tx)`
- `lat <- ncvar_get(ncin, "latitude")` and similar lines
- `dates <- as.Date(as.vector(time), origin = "1950-01-01")`

In this step the code reads the coordinate system and time range from the NetCDF files.

Step 8: Extract Climate Data

Code Reference: Lines 49-75

- # Prepare empty result dataframe

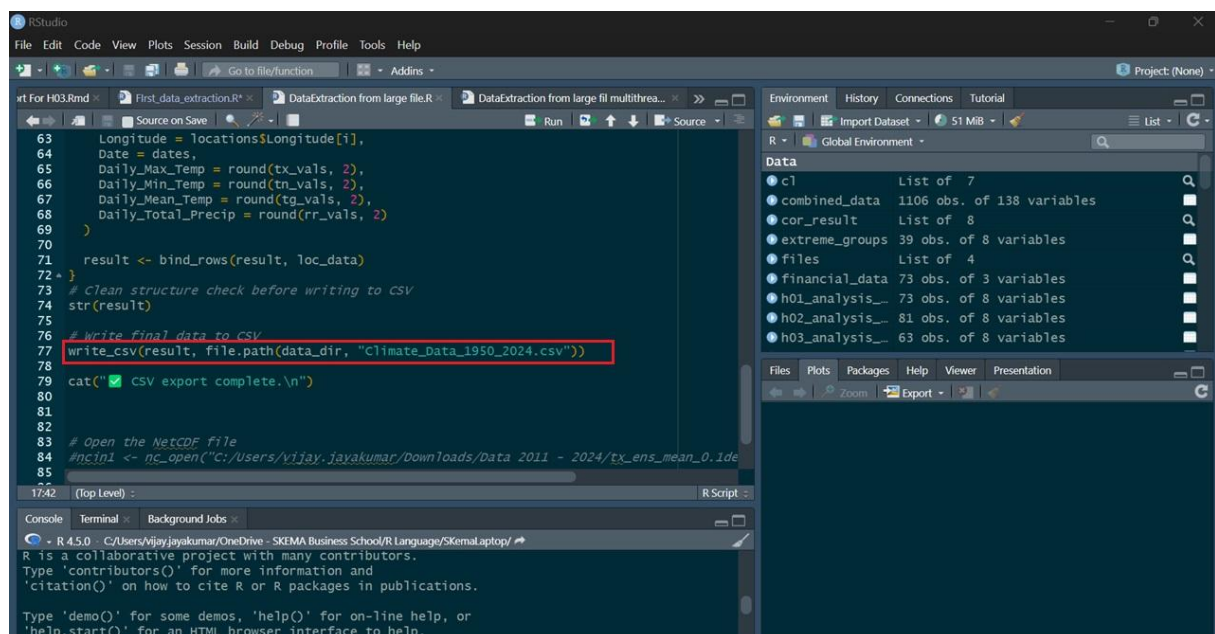
- # Iterate through locations
- for (i in 1:nrow(locations)) {...}
- Individual variable extraction with comments like # Read values from each NetCDF

In this step the code Loops through each location, finds the nearest data points, and extracts all climate variables.

Step 9: Quality Check and Export

Code Reference: Lines 76-81

- # Clean structure check before writing to CSV
- str(result)
- # Write final data to CSV
- write_csv(result, file.path(data_dir, "Climate_Data_1950_2024.csv"))
- The file name can be changed as per the users naming conventions.



Understanding the Output

The CSV file will contain:

- **Location:** Name that are assigned to each location
- **Latitude/Longitude:** Exact coordinates that are specified
- **Date:** Daily dates from the NetCDF files
- **Daily_Max_Temp:** Maximum temperature (°C)

- **Daily_Min_Temp:** Minimum temperature (°C)
- **Daily_Mean_Temp:** Average temperature (°C)
- **Daily_Total_Precip:** Total precipitation (mm)

To check climate data use below link (not the same data set –it is a reanalysis data set, not observations like E-OBS, hourly , not daily etc)

Check for Data Correctness R- Script VS QGIS

Spot check is performed for the data extracted using R- script vs data from QGIS for the location “L'ENCLOS DES BRAVES_43.860548_1.747921”. The data extracted using both the methods are the same as shown in the screen shot below.

Excel File - [Check QGIS vs R Script L'ENCLOS DES BRAVES_43.860548_1.747921.xlsx](#)

	A	B	C	D	E	F	G	H
	Location	Latitude						
	L'ENCLOS DES BRAVES	43.860548						
	R-Script Date	R-Script Daily_Max_Temp			Qgis Layer Name	Qgis Date	Qgis Daily_Max_Temp	Check
	01-01-2011	8.94			SAMPLE_1	01-01-2011	8.94	TRUE
	02-01-2011	2.95			SAMPLE_2	02-01-2011	2.95	TRUE
	03-01-2011	3.47			SAMPLE_3	03-01-2011	3.47	TRUE
	04-01-2011	5.53			SAMPLE_4	04-01-2011	5.53	TRUE
0	05-01-2011	11.29			SAMPLE_5	05-01-2011	11.29	TRUE
1	06-01-2011	14.71			SAMPLE_6	06-01-2011	14.71	TRUE
2	07-01-2011	14.4			SAMPLE_7	07-01-2011	14.4	TRUE
3	08-01-2011	14.19			SAMPLE_8	08-01-2011	14.19	TRUE
4	09-01-2011	9.02			SAMPLE_9	09-01-2011	9.02	TRUE
5	10-01-2011	7.7			SAMPLE_10	10-01-2011	7.7	TRUE
6	11-01-2011	11.99			SAMPLE_11	11-01-2011	11.99	TRUE
7	12-01-2011	11.61			SAMPLE_12	12-01-2011	11.61	TRUE
8	13-01-2011	14.1			SAMPLE_13	13-01-2011	14.1	TRUE
9	14-01-2011	12.94			SAMPLE_14	14-01-2011	12.94	TRUE
0	15-01-2011	14.46			SAMPLE_15	15-01-2011	14.46	TRUE
1	16-01-2011	14.8			SAMPLE_16	16-01-2011	14.8	TRUE
2	17-01-2011	14.24			SAMPLE_17	17-01-2011	14.24	TRUE
3	18-01-2011	13.58			SAMPLE_18	18-01-2011	13.58	TRUE
4	19-01-2011	9.96			SAMPLE_19	19-01-2011	9.96	TRUE
5	20-01-2011	7.21			SAMPLE_20	20-01-2011	7.21	TRUE